

Serena Huang

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EDUCATION

Massachusetts Institute of Technology

B.S., *Intended Double Major in Mathematics and Artificial Intelligence & Decision Making*

Cambridge, MA

Expected May 2030

Parkway Central High School

High School Diploma | GPA: 4.79/4.00 | ACT: 36/36 | SAT: 1560/1600

Chesterfield, MO

Expected May 2026

- **Honors & Awards:** NCWIT Aspirations in Computing National Winner (Top 40 of 2.7K+ Applicants), USACO Gold Division Competitor, National STEM Festival Finalist, 3x 1st Place Congressional App Challenge Winner, Girls Who Code Scholar

PROFESSIONAL EXPERIENCE

Bear Studios Consulting Group

Technology Consultant

St. Louis, MO

Dec 2023 - Apr 2026

- Developed digital placemaking mobile app for 50+ local businesses using React Native, Express.js, SQL, and AWS, and integrated map APIs, live video stream, and sensor data to enhance real-time user experience
- Overhauled website using Figma and React.js for celebrity gift bag client featuring 50+ brand partners and 100+ A-list events with 30M+ viewers, including the Grammys, Emmys, and Dancing with the Stars

Stanford University

Section Leader, Code in Place Program

Stanford, CA (Remote)

Apr 2025 - Jun 2025

- Selected from competitive global applicant pool to teach Python through Stanford's CS106A course to 40K+ learners
- Mentored 15 students in debugging and problem-solving during weekly discussion sections, and provided detailed code feedback

NASA & UT Austin's Center for Space Research

Aerospace Research Intern, STEM Enhancement in Earth Science (SEES) Program

Austin, TX

May 2024 - Aug 2024

- Applied k-means clustering and cosine similarity using NumPy and Scikit-Learn to compare 50+ terrestrial building materials, identifying Portland cement and brick as 93.5% matches to lunar basalt and regolith for Artemis mission planning
- Analyzed NASA astrophysical datasets to catalog 7.9K+ stars in Andromeda galaxy and generate velocity dispersion and temperature trend maps using NumPy and Matplotlib for galactic dynamics research
- Invited to present findings at American Geophysical Union's 2024 Annual Fall Meeting through Bright STaRs program

Ariel Premium Supply

Software Engineer Intern

St. Louis, MO

Jun 2023 - Aug 2023

- Applied transfer learning with EfficientNet in PyTorch to learn visually similarity embeddings of over 10K+ images collected from U.S. Patent & Trademark Office, detecting near-duplicate logos to flag potential trademark infringement
- Integrated learned representations into production logo search pipeline with OCR, achieving 85% accuracy on validation data

Software Engineer Intern

Jun 2022 - Aug 2022

- Built Selenium and Linux cron data pipeline scrape 600+ client records daily, saving 15+ hours per week and \$12K+ annually
- Developed spaCy natural language processing (NLP) pipeline to redact personally identifiable information from PDFs, text, and images, ensuring legal compliance

PERSONAL PROJECTS

AI Hydrological Modeling Research | PyTorch, Pandas, Matplotlib

github.com/serenahuang225/transformer-hydro-model

- Developed transformer-based rainfall prediction models with PyTorch and custom data loaders on CAMELS dataset covering 531 basins with 20-40 years of vegetation, topography, and soil data
- Benchmarked single-basin and multi-basin models, achieving Nash-Sutcliffe Efficiency score of 0.764 and outperforming LSTM baselines by 6%, classified as "Very Good" by hydrological modeling standards

Tunas Swim Meet Dashboard | Next.js, FastAPI, MCMC

github.com/serenahuang225/tunas-webapp-frontend

- Developed FastAPI backend hosted on Railway with Pydantic models and Markov Chain Monte Carlo (MCMC) simulator to analyze 800K+ race and relay outcomes for 1.2K+ USA Swimming clubs and 26K+ swimmers
- Built Next.js, TypeScript, and Recharts dashboard to visualize swimmer history, analytics, and relay optimization

Mock Operating System | C++

github.com/serenahuang225/mockos

- Implemented core file system operations (create, read, write, delete) using abstract base classes (AbstractFile) and concrete types (TextFile, ImageFile), and packaged operating system as static C++ library with CMake

Lively: Planner and Journal App | React Native, Express.js, Firebase, Figma

bubblystudios.com/lively

- Created Lively, a React Native productivity app with dynamic calendars, mood tracking, live chat, and social event planning, scaling to 1K+ iOS App Store downloads and 10K+ monthly interactions
- Implemented real-time notifications and friend synchronization via Firebase on an Express.js server hosted on Render, and built custom UI components (infinite pager, bottom sheet) with React Native Gesture Handler and Reanimated

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, C++, Java, SQL, JavaScript, Swift, HTML/CSS

Frameworks & Technologies: PyTorch, TensorFlow, Scikit-Learn, Keras, Pandas, NumPy, Matplotlib, React, Git, Jira

Certifications: Machine Learning & Intro to Statistical Learning (Stanford Online), CS50's Intro to Computer Science (HarvardX)